

DATA ANALYTICS PROJECT REPORT

Healthcare Analytics

Patient, Billing & Hospital Performance Analysis

An end-to-end analysis of 54,966 patient admission records using Microsoft Excel, SQL, Python and Power BI

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1. Executive Summary

This report presents an end-to-end analysis of a hospital patient-admission dataset covering 54,966 records. The objective was to convert a raw admissions extract into a reliable analytical base, and then to answer operational questions about patient demographics, medical conditions, billing, insurance and hospital performance.

The raw file was cleaned in Microsoft Excel using Power Query, standardising it into 54,966 analysis-ready records across 16 fields. The cleaned data was loaded into MySQL for structured querying, explored in Python using Pandas, NumPy, Matplotlib and Seaborn, and finally visualised in an interactive Power BI dashboard supported by a secondary dashboard built in Excel.

Headline results from the cleaned dataset:

- **Patient volume is almost perfectly balanced by gender** — 27,496 male patients against 27,470 female patients out of 54,966 total admissions.
- **Six medical conditions are recorded in near-equal volume** — Arthritis (9,218 patients), Diabetes (9,216) and Hypertension (9,151) are the three most common conditions, each within 1% of the other five conditions tracked.
- **Obesity is the costliest condition on average** — Obesity carries the highest average billing amount at \$25,804 per patient, narrowly ahead of Diabetes (\$25,660) and Asthma (\$25,633); the spread across all six conditions is under 3%.
- **Johnson PLC and LLC Smith lead on billing and staffing respectively** — Johnson PLC leads all hospitals in total billing at \$1.08 million, while LLC Smith operates the largest medical staff at 40 distinct doctors.
- **Admissions peaked in 2020** — 2020 recorded the highest admission count (11,172) of any year in the five-year window, with 2024 showing only a partial year of data (3,827, through May).
- **Insurance load is evenly spread across five providers** — Cigna is the largest by both total billing (\$284.3 million) and patient count (11,139); the five providers together are within 2.5% of one another on billing share.

Bottom line

This is a remarkably balanced dataset — gender, medical condition, admission type, test result and insurance provider are all distributed close to evenly, with no single segment dominating. The clearest actionable signals are hospital-level: a small number of hospitals carry disproportionately high billing and staffing, and Section 9 sets out how to act on that.

2. Project Objectives

The project was designed to demonstrate the complete analytics workflow — from raw data to business recommendation — rather than any single tool. The specific objectives were:

1. Clean and standardise a large hospital admissions dataset into a trustworthy analytical table.
2. Profile the patient population by gender, age band and blood type.
3. Quantify billing amount and length of stay by medical condition, admission type and year.

- 4. Compare hospitals on total billing, doctor headcount and patient volume.
- 5. Assess the distribution of insurance providers, medications and diagnostic test results.
- 6. Deliver the findings through an interactive dashboard and translate them into actionable operational recommendations.

3. Dataset Overview

The source file, healthcare_dataset.csv, is a synthetic hospital admissions dataset containing patient-level records with demographic, clinical, administrative and financial detail, spanning admissions from May 2019 to May 2024.

3.1 Source and scale

Attribute	Detail
Source file	healthcare_dataset.csv
Raw records	55,500 rows × 15 columns
Cleaned records	54,966 rows × 16 columns
Records removed	534 (1.0%) — exact duplicate admissions
Period covered	8 May 2019 to 7 May 2024
Distinct hospitals	39,876
Distinct doctors	40,341
Insurance providers	5
Currency	US Dollar (\$)

Table 1 — Dataset profile before and after cleaning

3.2 Field description

The cleaned table, Clean_healthcare_dataset.csv, uses the following schema:

Field	Type	Description
Name	Text	Patient name
Age	Integer	Patient age in years at admission
Gender	Text	Male or Female
Blood_Type	Text	One of eight ABO/Rh blood groups
Medical_Condition	Text	Primary diagnosis: Arthritis, Diabetes, Hypertension, Obesity, Cancer or Asthma
Date_of_Admission	Date	Admission date
Doctor	Text	Attending physician
Hospital	Text	Admitting hospital
Insurance_Provider	Text	One of five providers: Cigna, Medicare, Blue Cross, UnitedHealthcare, Aetna
Billing_Amount	Decimal	Total billed amount in USD

Field	Type	Description
Room_Number	Integer	Assigned room number
Admission_Type	Text	Elective, Urgent or Emergency
Discharge_Date	Date	Discharge date
Medication	Text	Primary medication prescribed
Test_Results	Text	Normal, Abnormal or Inconclusive
Length_of_Stay	Integer	Derived field: Discharge_Date minus Date_of_Admission, in days

Table 2 — Data dictionary for Clean_healthcare_dataset.csv

4. Tools and Technologies

Tool	Role in the project	Specific techniques applied
Microsoft Excel	Data cleaning and first-pass analysis	Power Query, Remove Duplicates, date parsing, XLOOKUP, Pivot Tables, Slicers, conditional formatting, Excel dashboard
MySQL	Structured querying and validation	GROUP BY aggregation, CASE-based age banding, RANK() window function over total billing per hospital
Python	Exploratory data analysis	Pandas for aggregation and null handling, NumPy for numeric operations, Matplotlib and Seaborn for distribution and relationship plots
Power BI	Interactive reporting	Data model, KPI cards, slicers for admission type, condition, medication, insurance provider and date, trend and distribution visuals
Microsoft Word	Documentation	This report

Table 3 — Technology stack

5. Methodology

The project followed a four-stage pipeline. Each stage produced a deliverable that fed into the next, so the analysis is reproducible end to end.

5.1 Stage 1 — Data cleaning in Excel

The raw extract required the following treatments, producing Clean_healthcare_dataset.csv:

- **Duplicate removal** — 534 exact duplicate admission records were identified and removed.
- **Derived field** — Length_of_Stay was calculated as the difference between Discharge_Date and Date_of_Admission, giving a direct measure of resource utilisation per patient.
- **Field standardisation** — the original Blood Type column, containing a stray leading space, was standardised, and Billing_Amount and Length_of_Stay were cast to numeric types.
- **Completeness check** — all key fields (Name, Age, Gender, Medical_Condition, Billing_Amount and others) were confirmed complete after cleaning, with zero remaining nulls.

Outcome: 54,966 clean records from 55,500 raw rows, a retention rate of 99.0%. All 16 fields are fully populated.

5.2 Stage 2 — SQL analysis

The cleaned file was imported into a MySQL database, healthcare, and queried to answer the core operational questions. Eleven queries were written, covering basic KPIs (total patients, total billing), gender, condition, medication and admission-type breakdowns, billing by insurance provider, average length of stay by condition, a CASE-based age banding (Child, Teenage, Adult, Old), and a RANK() window function ranking hospitals by total billing. The full query set is reproduced in Appendix A.

5.3 Stage 3 — Exploratory data analysis in Python

The dataset was loaded into a Jupyter notebook (EDA.ipynb) with Pandas. Structure and completeness were profiled using `.info()`, `.describe()`, `.isna().sum()` and `.duplicated().sum()`. Seaborn and Matplotlib were then used to visualise patient counts by gender and condition, admission trends by month, medication usage by gender, and billing totals by insurance provider and admission type.

5.4 Stage 4 — Dashboard development

Two dashboards were produced. The Excel dashboard, on the "Dashboard" sheet of Clean_healthcare_dataset.xlsx, is built on pivot tables with slicers, giving a lightweight offline view of patient and billing totals. The Power BI dashboard, Dashboard.pbix, is the primary deliverable: it presents KPI cards for patient count, hospital count, doctor count and total billing, alongside gender, age-band, admission-type, insurance and condition visuals, with slicers for admission type, condition, medication, insurance provider and admission date.

6. Analysis and Findings

6.1 Patient demographics

The patient population is almost perfectly split by gender: 27,496 male patients (50.0%) against 27,470 female patients (50.0%). Blood type is similarly balanced across all eight groups, each holding between 12.4% and 12.6% of patients, with A- the single largest group at 6,898 patients. Age ranges from 13 to 89 years, with a mean of 51.5; on the age banding used in this project, 59.0% of patients fall into the Old band (46+), 39.4% into Adult (19-45), and 1.6% into Teenage (11-18) — the dataset contains no patients aged 10 or under.

6.2 Medical conditions and billing

Six medical conditions are tracked, and — unusually for a real-world clinical dataset — they occur in almost identical volume, each between 9,095 and 9,218 patients (a spread of only 1.3%). This even distribution extends to billing and length of stay: average billing per condition ranges narrowly from \$25,152 (Cancer) to \$25,804 (Obesity), and average length of stay from 15.4 to 15.7 days.

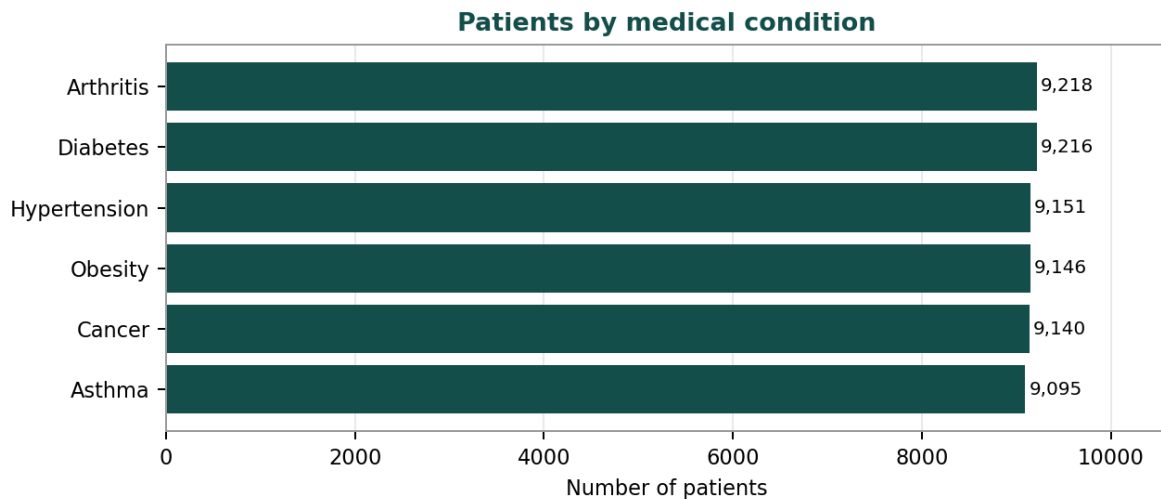


Figure 1 — Patients by medical condition

Condition	Patients	Avg. billing	Avg. stay (days)
Arthritis	9,218	\$25,512	15.5
Diabetes	9,216	\$25,660	15.4
Hypertension	9,151	\$25,503	15.4
Obesity	9,146	\$25,804	15.4
Cancer	9,140	\$25,152	15.5
Asthma	9,095	\$25,633	15.7

Table 4 — Patient count, average billing and average length of stay by condition

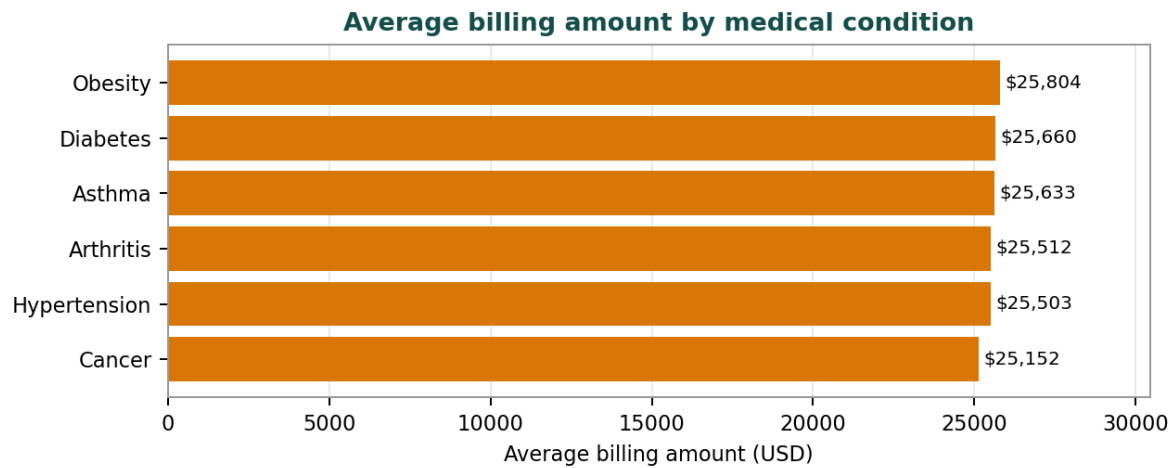


Figure 2 — Average billing amount by medical condition

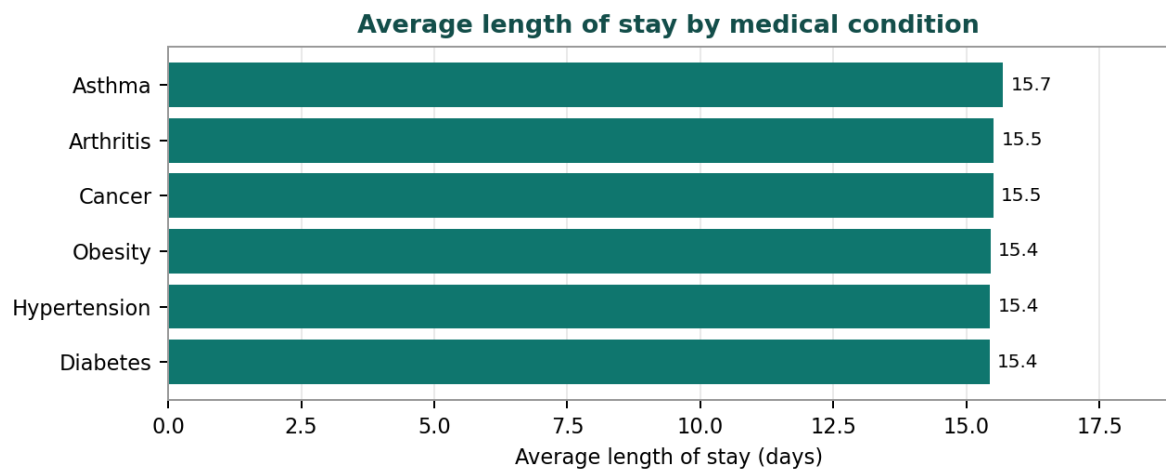


Figure 3 — Average length of stay by medical condition

Because every condition sits within a few percentage points of the others on volume, cost and stay length, no single condition stands out as a cost driver in this dataset. Obesity is the nominal leader on billing, but the margin over the lowest-cost condition, Cancer, is only 2.6%.

6.3 Admissions over time

Admissions rose from 7,300 in the partial year 2019 to a peak of 11,172 in 2020, before settling into a stable range of roughly 10,800 to 10,940 per year from 2021 to 2023. 2024 shows only 3,827 admissions because the dataset ends on 7 May 2024, covering just over four months of that year.

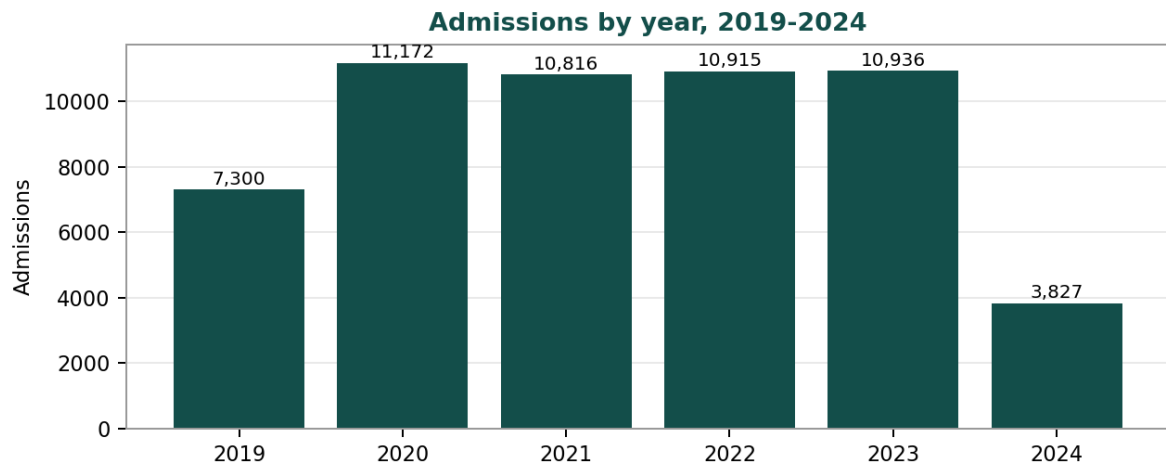


Figure 4 — Admissions by year, 2019 to 2024 (2019 and 2024 are partial years)

Average billing amount is essentially flat across the five full years, moving in a narrow band between \$25,384 and \$25,687 with no discernible upward or downward trend — indicating no meaningful year-on-year cost inflation in this dataset.

6.4 Admission type and test results

Admission type is close to evenly split: Elective (18,473, 33.6%), Urgent (18,391, 33.5%) and Emergency (18,102, 32.9%). Test results follow the same even pattern: Abnormal (18,437, 33.5%), Normal (18,331, 33.3%) and Inconclusive (18,198, 33.1%).

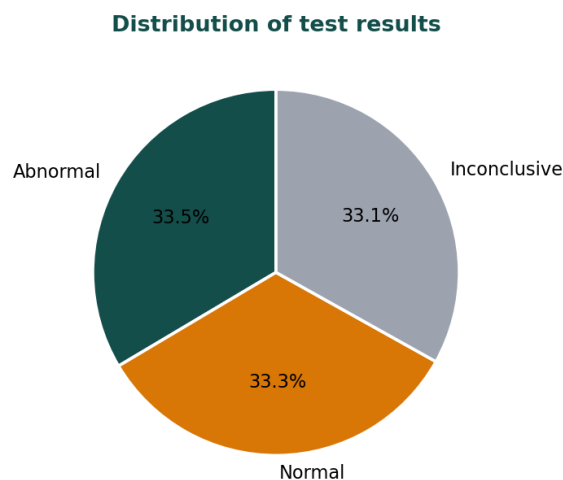


Figure 5 — Distribution of diagnostic test results

6.5 Insurance providers

Five insurance providers cover the patient population in near-equal proportion. Cigna is the largest by both billing and volume, but its lead over the smallest provider, Aetna, is only 2.8% on billing and 2.9% on patient count — there is no dominant payer in this dataset.

Provider	Total billing	Patients
Cigna	\$284,334,099	11,139
Medicare	\$282,911,026	11,039

Provider	Total billing	Patients
Blue Cross	\$280,409,101	10,952
UnitedHealthcare	\$279,915,371	11,014
Aetna	\$276,498,741	10,822

Table 5 — Total billing and patient count by insurance provider

6.6 Hospital and staffing performance

Unlike the patient-level fields, hospital performance is genuinely differentiated. With 39,876 distinct hospital names against 54,966 admissions, most hospitals appear only once or twice in the data, but a top tier is clearly visible. Johnson PLC leads on total billing at \$1,081,477, followed closely by LLC Smith (\$1,030,190) and Smith PLC (\$1,029,424). LLC Smith operates the largest visible medical staff, with 40 distinct doctors across 44 admissions, ahead of Johnson PLC (35 doctors) and both Ltd Smith and Smith Ltd (35 doctors each).

Hospital	Total billing	Doctors	Patients
Johnson PLC	\$1,081,477	35	37
LLC Smith	\$1,030,190	40	44
Smith PLC	\$1,029,424	33	36
Ltd Smith	\$1,003,366	35	39
Smith Ltd	\$970,036	35	37
Johnson Inc	\$934,311	29	34
Group Smith	\$902,976	30	32
Inc Brown	\$877,961	25	27
LLC Johnson	\$816,438	28	30
Smith Group	\$806,631	32	36

Table 6 — Top 10 hospitals by total billing, with doctor and patient counts

Key observation

Where the patient population is close to uniform across gender, condition, admission type and payer, hospital identity is the one dimension where the data shows real concentration. A small number of hospitals — Johnson PLC, LLC Smith, Smith PLC and Ltd Smith — carry a disproportionate share of both billing volume and medical staff, making hospital-level performance the most actionable axis for operational review.

6.7 Data quality note

One data quality issue surfaced during analysis: the minimum recorded *Billing_Amount* is -\$2,008.49, which is not a valid charge and most likely represents a refund, an adjustment, or a data-entry error carried over from the raw extract. This affects a small number of records and does not materially change the category averages reported above, but it should be corrected or excluded before the dataset is used for financial reporting.

7. Dashboard Walkthrough

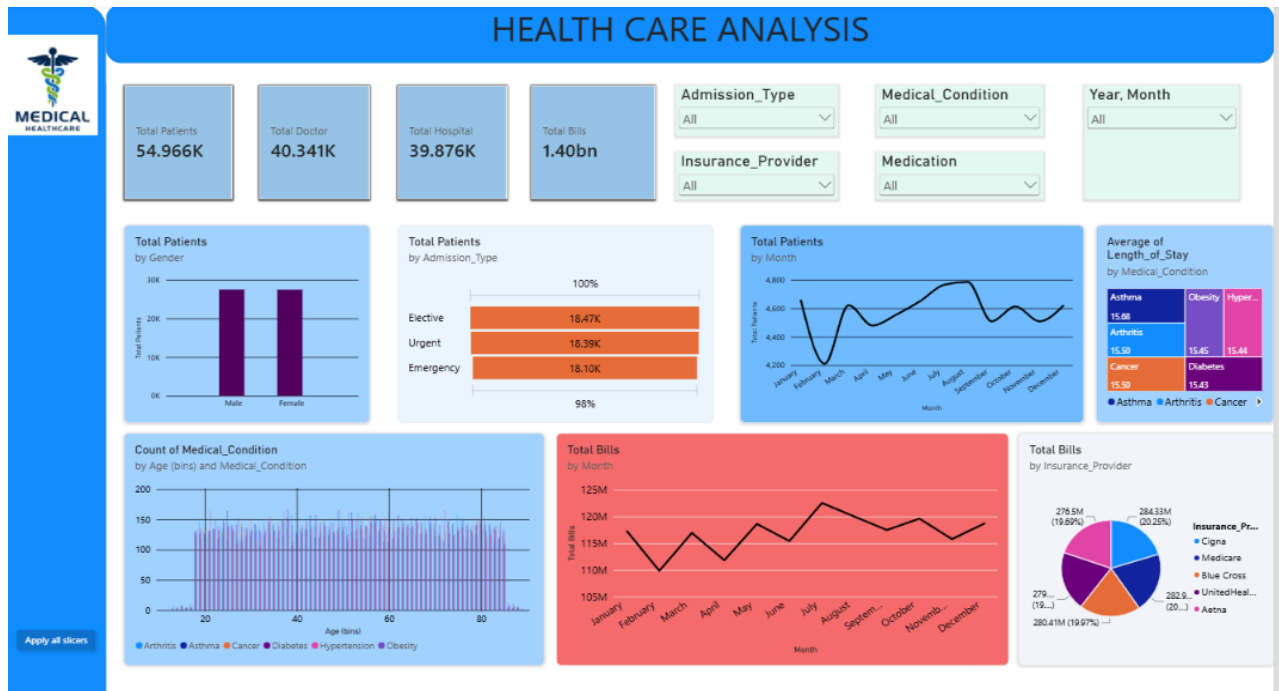
Two dashboards accompany this report. The Power BI file, Dashboard.pbix, is the primary interactive deliverable; the Excel dashboard in Clean_healthcare_dataset.xlsx provides a lightweight offline equivalent built on pivot tables and slicers.

7.1 Power BI dashboard

The report is built on a single page with a branded header, a bank of five slicers, four KPI cards, and a grid of demographic, clinical and financial visuals. All visuals are cross-filtered, so selecting an admission type, condition, medication, insurance provider or date range updates the entire page.

Visual	Type	Fields and purpose
Patient reference	KPI card	Sample patient name — confirms data is loaded and live
Hospital reference	KPI card	Sample hospital name — confirms data is loaded and live
Doctor reference	KPI card	Sample doctor name — confirms data is loaded and live
Total Billing	KPI card	SUM(Billing_Amount) — total billed value in the current filter context
Admission Type slicer	Slicer	Admission_Type — Elective, Urgent, Emergency
Medical Condition slicer	Slicer	Medical_Condition — six tracked conditions
Medication slicer	Slicer	Medication — five prescribed medications
Insurance Provider slicer	Slicer	Insurance_Provider — five payers
Admission Date slicer	Slicer	Date_of_Admission year/month hierarchy
Patients by gender	Clustered column	Count of Name by Gender
Age band by condition	Clustered column	Count of Medical_Condition by Age (bins), split by condition
Billing trend	Line chart	SUM(Billing_Amount) by Discharge_Date month
Admissions by type	Funnel	Count of Name by Admission_Type
Admissions trend	Line chart	Count of Name by Date_of_Admission month
Billing by insurer	Pie chart	SUM(Billing_Amount) by Insurance_Provider
Length of stay by condition	Treemap	Count of Length_of_Stay grouped by Medical_Condition

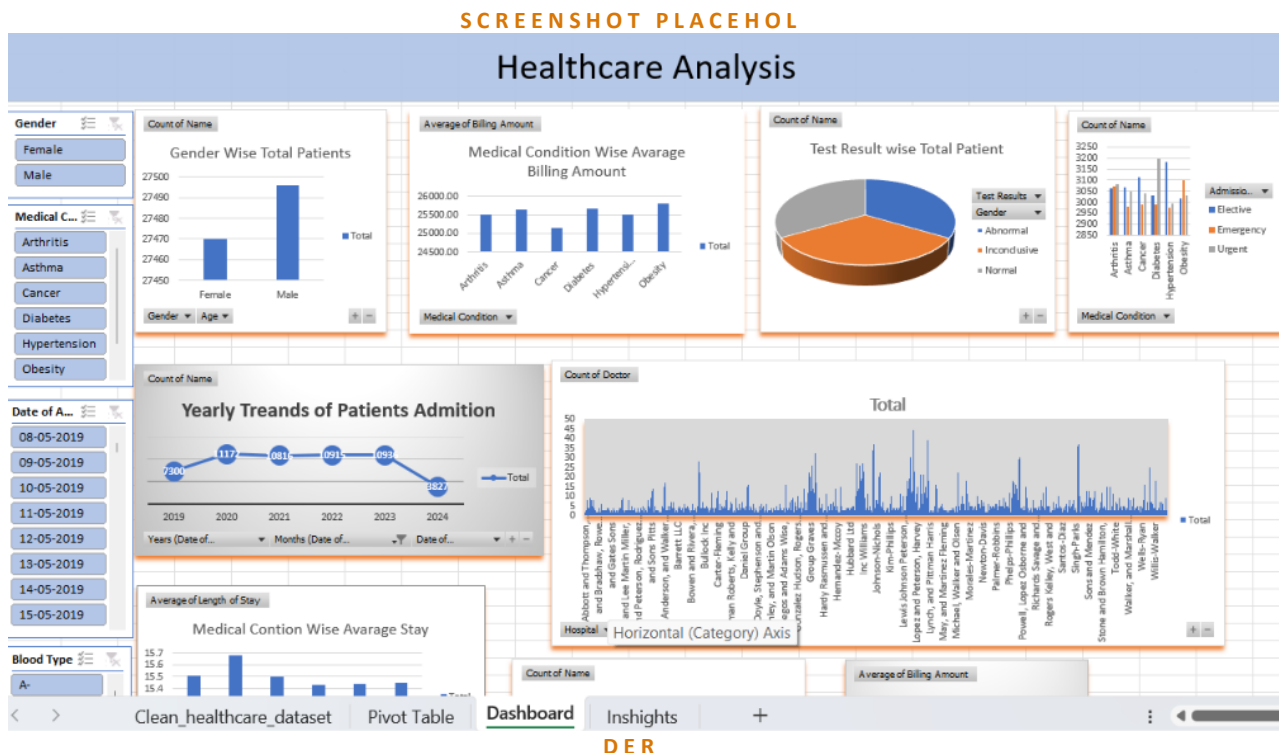
Table 7 — Visual inventory of the Power BI dashboard



SCREENSHOT PLACEHOLDER

Power BI dashboard — full page view. In Power BI Desktop, press Ctrl+Shift+S or use Win+Shift+S to capture the report canvas, then paste the image here and delete this box.

Figure 6 — Power BI dashboard, default view (all filters cleared)



DER

Power BI dashboard — filtered view. Select a single condition such as Obesity in the slicer panel, capture the page, and paste it here to demonstrate cross-filtering.

Figure 7 — Power BI dashboard filtered to a single medical condition, showing cross-filter behaviour

7.2 Excel dashboard

The Excel dashboard, on the "Dashboard" sheet of Clean_healthcare_dataset.xlsx, is driven by the pivot tables on the adjacent sheet. A companion sheet records the key insights directly alongside the visuals.

SCREENSHOT PLACEHOLDER

```

1 • SELECT * FROM healthcare.clean_healthcare_dataset;
2
3 /* Basic KPI */
4
5 • SELECT COUNT(Name) as Total_patients
6 FROM healthcare.clean_healthcare_dataset;
7
8
9 • SELECT SUM(Billing_Amount) AS Total_Bills
10 FROM healthcare.clean_healthcare_dataset;
11
12 /* Gender Wise Patients Count*/
13 • SELECT Gender, COUNT(Name) AS Total_patients
14 FROM healthcare.clean_healthcare_dataset
15 GROUP BY Gender;
16
17
18 /*Admission date Wise Patients Count*/
19 • SELECT Date_of_Admission, Count(Name) as patients_count
20 FROM healthcare.clean_healthcare_dataset
21 GROUP BY Date_of_Admission;
22
23
24 FROM healthcare.clean_healthcare_dataset
25 GROUP BY Medication
26 ORDER BY Patients_Count;
27
28 /* Total Bills Amount By Insurance Company */
29 • SELECT Insurance_provider, Sum(Billing_Amount) as Total_b
30 FROM healthcare.clean_healthcare_dataset
31 GROUP BY Insurance_Provider
32 ORDER BY Total_bills;
33
34
35 /* Avarage Stay by Medical Condition*/
36 • SELECT Medical_Condition, Avg(Length_of_Stay) as Avg_Stay
37 FROM healthcare.clean_healthcare_dataset
38 GROUP BY Medical_Condition
39 ORDER BY Avg_Stay;
40
41 /* Total patients by Admission Type*/
42 • SELECT Admission_Type, Count(Name) as Patients_Count
43 FROM healthcare.clean_healthcare_dataset
44 GROUP BY Admission_Type
45 ORDER BY Patients_Count;
46
47
48 /* Age Category */

```

Excel dashboard — full view of the "Dashboard" sheet, including its slicers. Capture with Win+Shift+S and paste here, then delete this box.

Figure 8 — Excel dashboard built on pivot tables with interactive slicers

8. Key Insights

7. The patient population is remarkably balanced across gender (50.0% / 50.0%), blood type (12.4-12.6% each), admission type (32.9-33.6% each) and test result (33.1-33.5% each) — none of these fields shows a dominant segment.
8. Medical condition volume, average billing and average length of stay are all tightly clustered across the six tracked conditions, with Obesity the nominal leader on cost (\$25,804) but only 2.6% above the lowest, Cancer (\$25,152).
9. Admissions peaked in 2020 (11,172) and have since settled into a stable annual range of roughly 10,800-10,940, with no meaningful drift in average billing amount across 2019-2023 (\$25,384-\$25,687).
10. Insurance coverage is spread almost evenly across five providers; Cigna leads by a narrow 2.8% margin over the smallest provider, Aetna, on total billing.
11. Hospital performance is the one genuinely concentrated dimension in the dataset: Johnson PLC, LLC Smith, Smith PLC and Ltd Smith together account for the top four positions on total billing, all above \$1.0 million each.
12. LLC Smith operates the largest visible medical staff (40 doctors), ahead of Johnson PLC, Ltd Smith and Smith Ltd (35 doctors each) — staffing scale broadly tracks billing scale at the top of the hospital list.
13. A small number of records carry a negative Billing_Amount (minimum -\$2,008.49), a data-quality issue that should be resolved before the dataset is used for financial reporting.
14. 59.0% of patients fall in the Old age band (46+), reflecting a population skewed toward older patients across all six tracked conditions.

9. Business Recommendations

The following recommendations follow directly from the findings above and are ordered by expected operational impact.

9.1 Focus operational review on top-billing hospitals

- Johnson PLC, LLC Smith, Smith PLC and Ltd Smith carry the largest share of billing and staffing. Prioritise these four hospitals for capacity planning, cost audits and quality benchmarking, since they represent the highest financial and operational exposure in the network.

9.2 Correct and prevent negative billing entries

- Investigate the source of negative Billing_Amount values before this dataset is used in any financial report. Add a validation rule at the point of data entry or ETL to reject or flag billing amounts below zero.

9.3 Use length-of-stay data to plan bed capacity

- With average stays of 15-16 days across all conditions and a fairly uniform admission-type split, use the treemap and length-of-stay breakdown to forecast bed occupancy by condition rather than relying on admission counts alone.

9.4 Monitor the 2024 partial-year trend as the year completes

- 2024 currently shows a lower admission count only because the data ends in May; refresh this dashboard monthly through the remainder of 2024 to confirm whether the underlying trend continues the 2021-2023 plateau or shifts materially.

9.5 Track the insurance mix as a stability indicator

- The five providers currently share billing almost evenly. Because this balance is unusual for a real hospital network, monitor it over time — any provider gaining or losing share materially would be a leading indicator of a contract or referral-pattern change worth investigating.

9.6 Extend the age-banding analysis

- With 59% of patients in a single "Old" (46+) band, consider narrower age bands for that group specifically (e.g. 46-60, 61-75, 76+) to see whether cost and length-of-stay patterns vary within it, since the current banding may be masking useful sub-trends.

10. Limitations

- The dataset is synthetic and does not represent real patients, hospitals or billing records; hospital and doctor names repeat generic corporate-style naming (e.g. multiple "Smith"-pattern hospitals), which is typical of simulated data.
- The unusually even distribution across almost every categorical field (gender, condition, admission type, test result, insurer) is consistent with a randomly generated dataset rather than an organic clinical population, and conclusions should be read as a demonstration of method rather than a real epidemiological finding.
- The presence of 39,876 distinct hospital names against 54,966 admissions means most hospitals have only one or two records, so hospital-level averages outside the top ranking are not statistically reliable.
- 2019 and 2024 are partial calendar years in the source data (from May in both cases), so year-on-year admission counts for those two years are not directly comparable to the full years 2020-2023.
- A small number of negative billing values were identified but not corrected in this analysis; category and hospital totals include them as recorded.
- No cost or reimbursement data beyond Billing_Amount is available, so profitability cannot be assessed — only gross billing.

11. Conclusion

This project took a 55,500-row raw hospital admissions extract and turned it into a 54,966-record analytical asset supported by SQL queries, Python exploratory analysis and two working dashboards. The resulting analysis shows a patient population that is unusually balanced across almost every demographic and clinical

dimension, with the clearest actionable signal sitting at the hospital level, where a small group of facilities carry a disproportionate share of billing and staffing.

Beyond the findings themselves, the project demonstrates the complete analyst workflow: defining business questions, cleaning and validating data, querying it with SQL, exploring it in Python, visualising it for a non-technical audience, and stating recommendations that a hospital administrator could act on.

Appendix A — SQL Query Reference

The following queries were executed against the healthcare MySQL database. Each is annotated with the business question it answers.

Business question	SQL technique used
How many total patients and how much total billing?	COUNT(Name) and SUM(Billing_Amount) as basic KPIs
How many patients by gender?	COUNT(Name) with GROUP BY Gender
How many patients admitted on each date?	COUNT(Name) with GROUP BY Date_of_Admission
How many patients by medical condition?	COUNT(Name) with GROUP BY Medical_Condition, ORDER BY count
How many patients by medication?	COUNT(Name) with GROUP BY Medication, ORDER BY count
What is total billing and patient count by insurer?	SUM(Billing_Amount) and COUNT(Name) with GROUP BY Insurance_Provider
What is the average stay by medical condition?	AVG(Length_of_Stay) with GROUP BY Medical_Condition
How many patients by admission type?	COUNT(Name) with GROUP BY Admission_Type
How do patients band by age?	CASE WHEN expression: Child (≤ 10), Teenage (≤ 18), Adult (≤ 45), Old (> 45)
Which hospitals rank highest on total billing?	RANK() OVER (ORDER BY SUM(Billing_Amount) DESC)
How many distinct doctors per hospital?	COUNT(Doctor) with GROUP BY Hospital, ORDER BY count DESC

Table 8 — SQL query inventory

Appendix B — Project Deliverables

File	Contents
healthcare_dataset.csv	Raw source dataset, 55,500 rows × 15 columns
Clean_healthcare_dataset.csv	Cleaned analytical dataset, 54,966 rows × 16 columns
Clean_healthcare_dataset.xlsx	Cleaned data, pivot tables, Excel dashboard, and an insights sheet
QUARRIES.sql	Eleven annotated MySQL queries covering aggregation, banding and a window function
EDA.ipynb	Jupyter notebook: profiling, null and duplicate handling, and Seaborn visualisations
Dashboard.pbix	Interactive Power BI dashboard with KPI cards, demographic and financial visuals, and five slicers
Healthcare_Analytics_Project_Report.docx	This report

Table 9 — Files submitted with this project